2 b) Induktionsbevis

Base case:

Antag att $n = 1$.

$$\sum_{r=1}^{2(1)-1} \frac{1}{r(r+1)} = \frac{1}{2n} \Leftrightarrow \sum_{r=1}^{1} \frac{1}{1(1+1)} = \frac{1}{2},$$ så om $n = 1$ – så stämmer det.

$$\boxed{\sum_{r=k+1}^{2k+1} \frac{1}{r(r+1)} = \frac{1}{2(k+1)}}$$

Induktion Hypothesis:

Antag att $n = k$.

$$\sum_{r=k}^{1} \frac{1}{k(k+1)} = \frac{1}{2k}$$

Induktionsteps:

Antag att $n = k+1$.

$$\sum_{r=k+1}^{2(k+1)-1} \frac{1}{r(r+1)} = \frac{1}{2(k+1)} \Leftrightarrow \sum_{r=k+1}^{2k+1} \frac{1}{r(r+1)} = \frac{1}{2(k+1)}$$

$$\sum_{r=k}^{2k-1} \frac{1}{r(r+1)} = \frac{1}{2k} = \sum_{r=k}^{2k-1} \frac{1}{2k} + \frac{1}{(2k)(2k+1)}$$

$$\frac{1(2k+1)}{2k(2k+1)} + \frac{1}{2k(2k+1)} = \frac{2k+1}{2k(2k+1)} + \frac{1}{2k(2k+1)} = \frac{2k+2}{2k(2k+1)} = \frac{\cancel{2}(k+1)}{\cancel{2}k(2k+1)}$$

$$= \frac{k+1}{k(2k+1)} = \frac{1}{(2k+1)} = \boxed{\frac{1}{2(k+1)}}$$

SVAR: